TimeFlight

When flights don't move, your time and money do.

Usually in the wrong direction…

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Executive Summary

Flight delays create upwards of 33 billion in annual losses across the U.S. aviation system and remain difficult to anticipate at scale. TimeFlight is a regional forecasting framework that converts historical BTS delay data into forward-looking monthly risk signals. The models surface predictable seasonal patterns, highlight structural shifts in delay behavior, and generate practical forecasts that airlines, airports, and travel platforms can use for staffing, scheduling, and contingency planning. The result is a lightweight, data-driven tool that improves operational awareness and reduces uncertainty around future delay pressure.

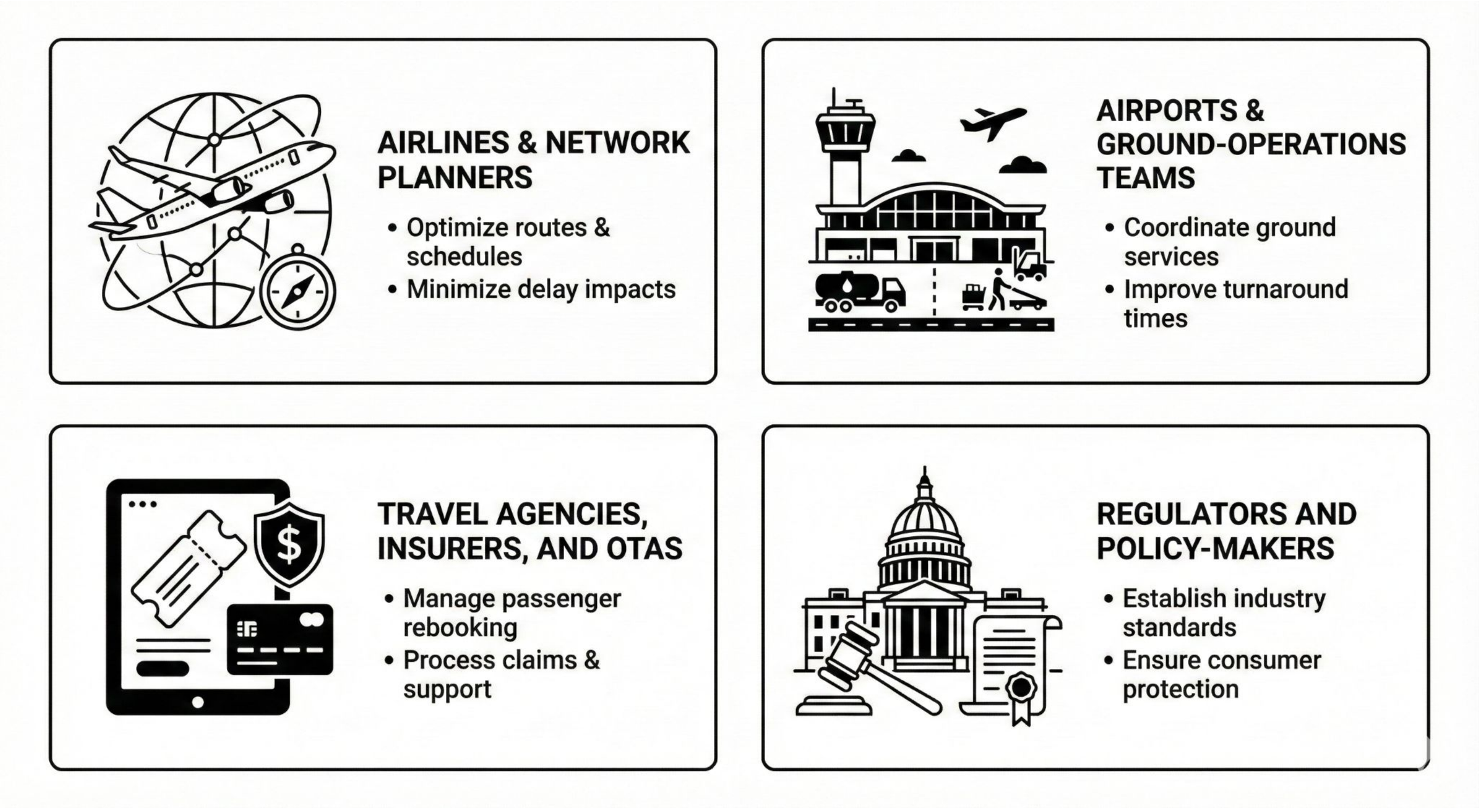
https://www.vaisala.com/en/press-releases/2025-01/top-airports-and-towns-us-struck-lightning-2024-vaisala-xweather-annual-lightning-report-here

Problem Context

$33 billion in Annual Losses!

https://www.vaisala.com/en/press-releases/2025-01/top-airports-and-towns-us-struck-lightning-2024-vaisala-xweather-annual-lightning-report-here

TimeFlight Forecasting benefits



Inputs - Data Ingestion

Source: Airline On-Time Performance Delay Cause dataset from the U.S. Bureau of Transportation Statistics (BTS)

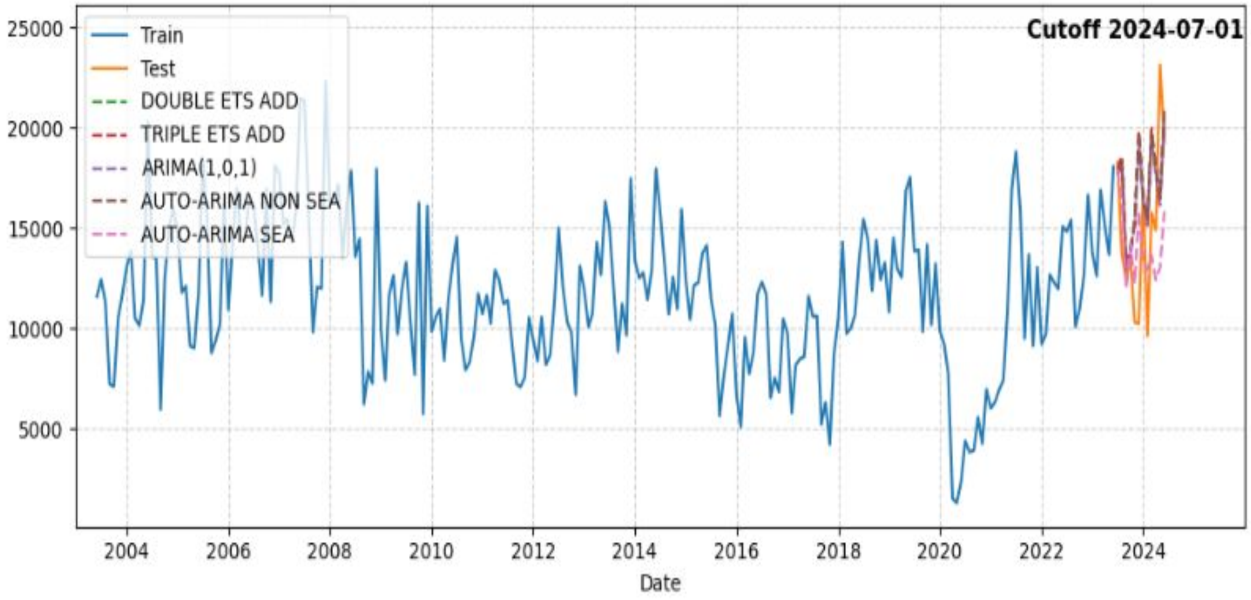
Time period: 2004 - 2024 (21 years)

The dataset contains 407,721 rows and 21 variables, Each row represents a single airport in a given month, and the variables cover monthly arrival statistics, delay counts, delay causes, and flight activity across all U.S. airports.

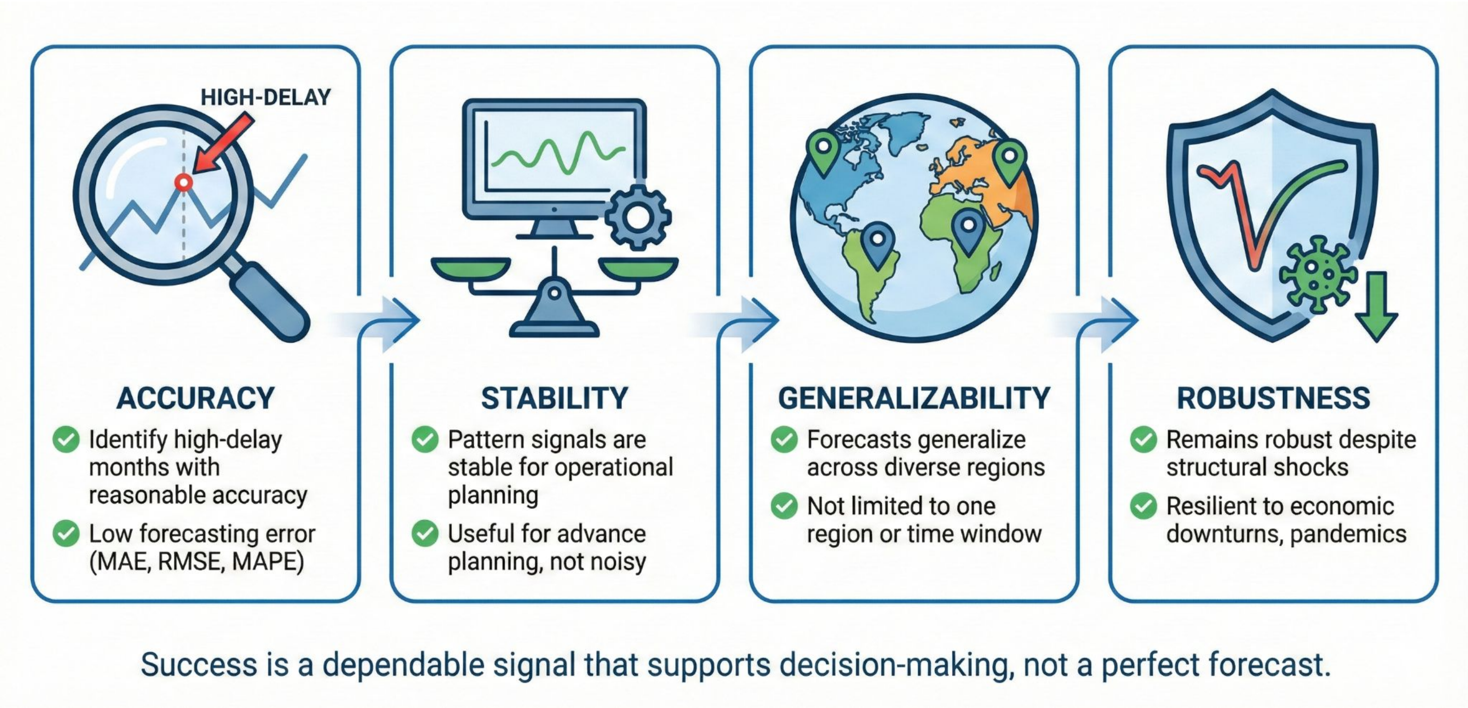
- **Time indicators:** `year`, `month`
- **Airport identifiers:** `airport`, `airport_name`
- **Flight volume metrics:** `arr_flights`
- **Delay occurrence:** `arr_del15`, `arr_cancelled`, `arr_diverted`
- **Delay causes:** `carrier_ct`, `weather_ct`, `nas_ct`, `security_ct`, `late_aircraft_ct`
- **Delay minutes:** `carrier_delay`, `weather_delay`, `nas_delay`, `security_delay`, `late_aircraft_delay`

Outputs - Delay Forecasting

- Monthly delay-risk forecasts per region (next 12 months)
- Method: univariate time-series models: ETS, ARIMA / seasonal ARIMA
- Output metrics: expected delay-counts, plus uncertainty band (for risk planning)



Success Criteria



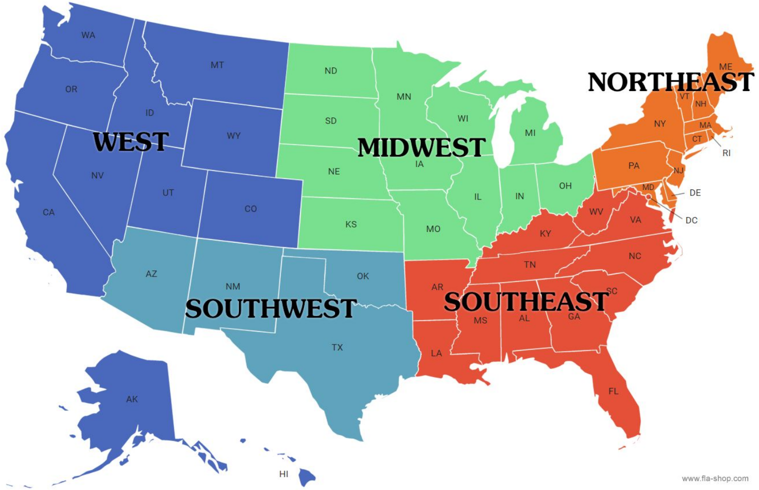
Understanding the Data

Data Cleaning and Preparation

- Defining 5 regions to bucket all airports
- Mapping airport codes manually to the 5 regions within our dataframe

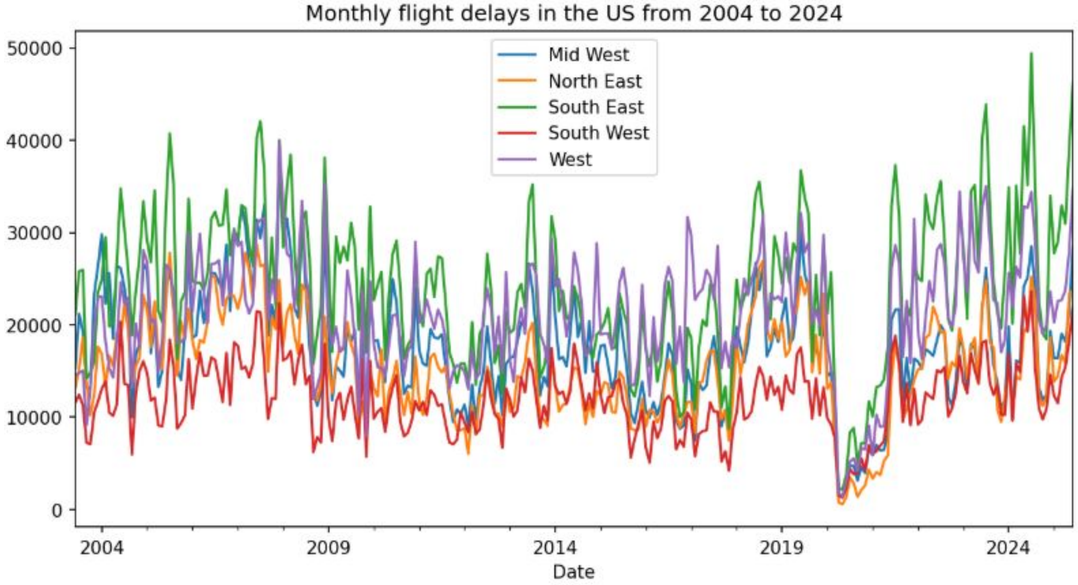
```
regions = {
"North East" : {"WVL", "SFM", "RKD", "PWM", "PQI", "PNN", "PG1", "OWK", "OLD", "MVM", "
"Mid West" : {"ORD", "MDW", "BLV", "IND", "SBN", "FWA", "DSM", "CID", "ICT", "MHK", "DT
"West" : {'LAX', 'SFO', 'SAN', 'OAK', 'SJC', 'LAS', 'PHX', 'SLC', 'DEN', 'SEA', 'PDX',
"South West" : {'PHX', 'ABQ', 'OKC', 'DFW', 'IAH', 'HOU', 'AUS'},
"South East" : {"ABY", "AEX", "AGS", "APF", "ATL", "AVL", "BFM", "BHM", "BNA", "BQK", "
}
```

- Converting year and month into proper datetime format so models can interpret it chronologically
- Delays are defined in the dataset as >15 minutes
- No missing values - phew!

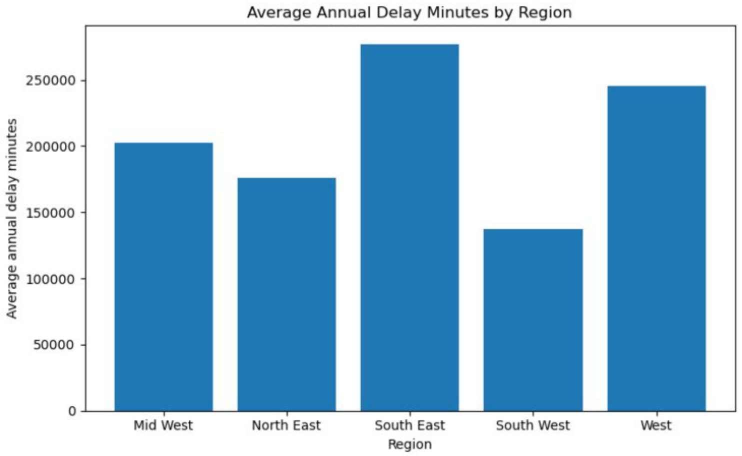


After cleaning and aggregation, we use a region-level monthly dataset where each row represents one of five U.S. regions and its total arrival delay minutes for a given month.

EDA - The Data Story

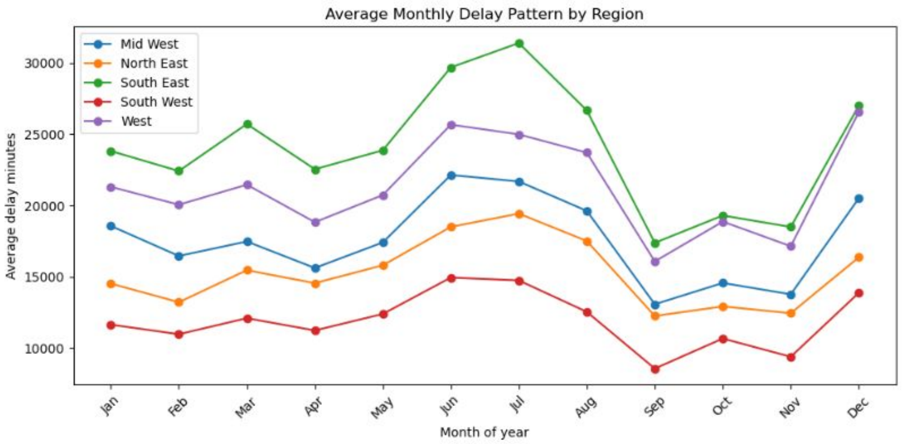


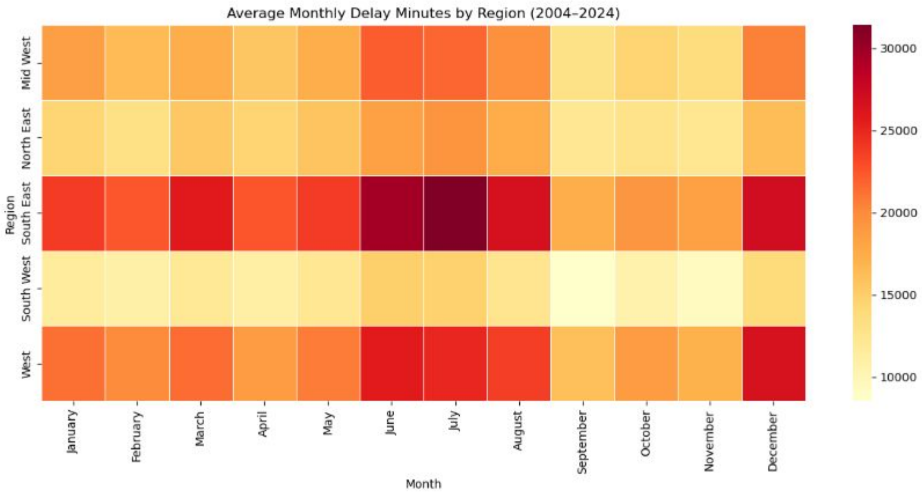
Delay counts vary widely by region.
Spikes in the middle, end of every year?



The South East has the highest average annual delays, followed by the West. The Mid West and North East sit in the middle, and the South West has the lowest overall delay burden.

Seasonality Impacts





Strong month-of-year seasonality in several regions. Heatmap reveals repeating seasonal stripes.

- Winter peaks in Northeast & Midwest
- Summer storm season impacts Southeast
- Identifies structural changes and anomaly periods

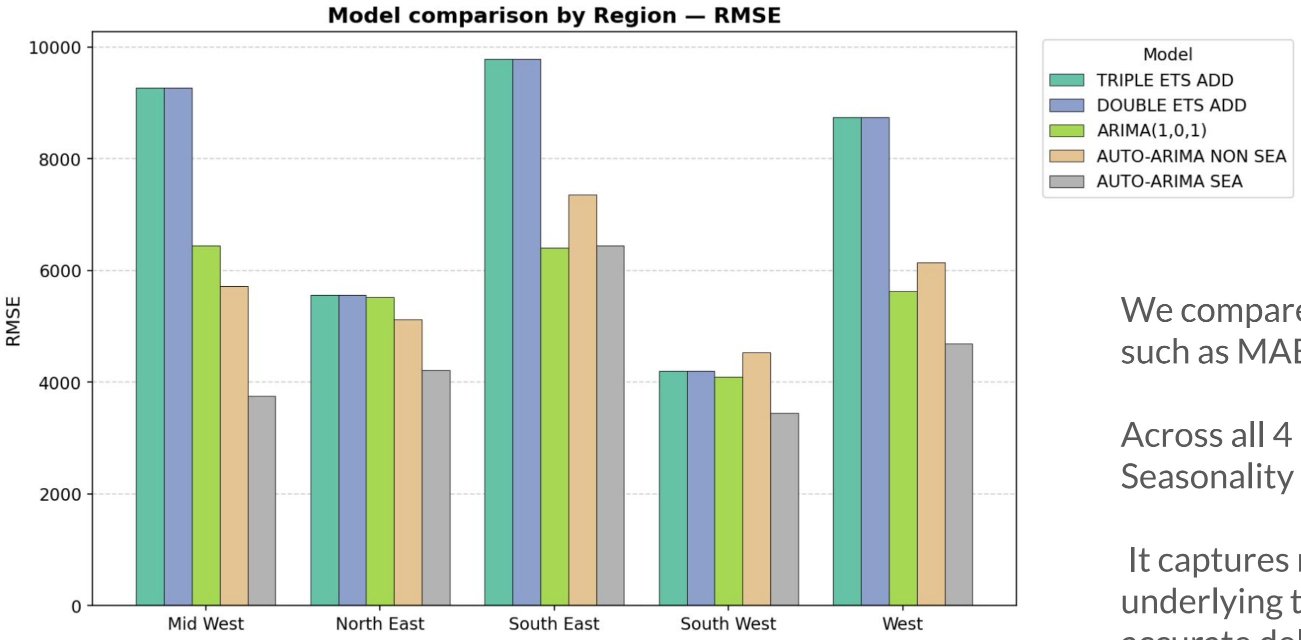
Time Series Modelling Approach

- Double ETS $\rightarrow$ Forecast = Level+Trend
- Triple ETS $\rightarrow$ Forecast = Level + Trend + Seasonality
- ARIMA (1,0,1) $\rightarrow$ Future Value=a $\cdot$ Past Value+b $\cdot$ Past Error
- Auto ARIMA (non-seasonal) $\rightarrow$ Future Value=f(Past Values, Past Errors)
- Auto ARIMA (seasonal) $\rightarrow$ Future Value=f(Past Values, Past Errors, Seasonal Patterns)

Cutoff Based Forecasting Design

2009-01-01	2016-01-01	2020-01-01	2024-07-01
Early-period data split	Mid-period operational change	COVID-19 structural break	Most recent high-volatility period
Pre-2010 delay patterns	Clearer post-2016 trends	Tests post-shock model behavior	Smallest training window, most realistic

Model Evaluation & Accuracy Metrics

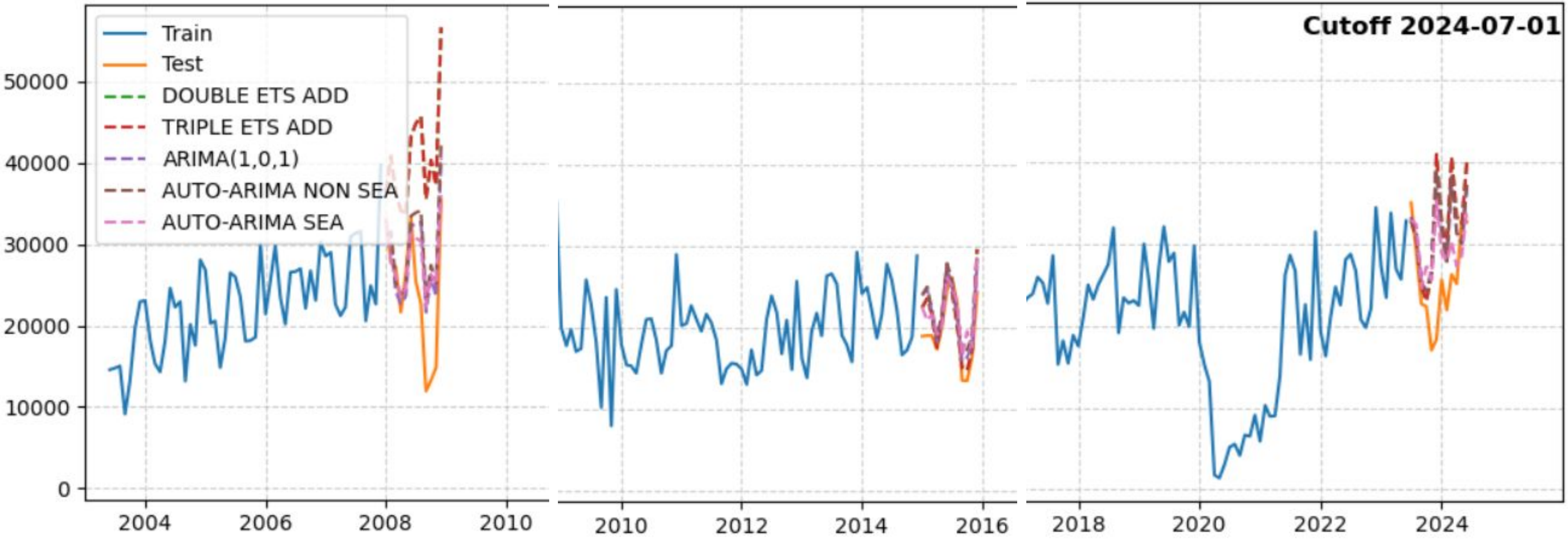


We compared the 5 models through metrics such as MAE, R2, MSE, and ME.

Across all 4 metrics, Auto-Arima with Seasonality stands out the best.

It captures recurring seasonality and the underlying trends to generate the most accurate delay predictions.

Model Evaluation & Accuracy Metrics



Effectiveness & Limitations

Effectiveness

Working Efficiency

Forecasts are not perfect, but they consistently identify high-delay periods in advance. That alone is valuable

- Forecasts reliably indicate high-delay months
- Useful directional signals for planning
- Stable performance across multiple cutoffs

Limitations

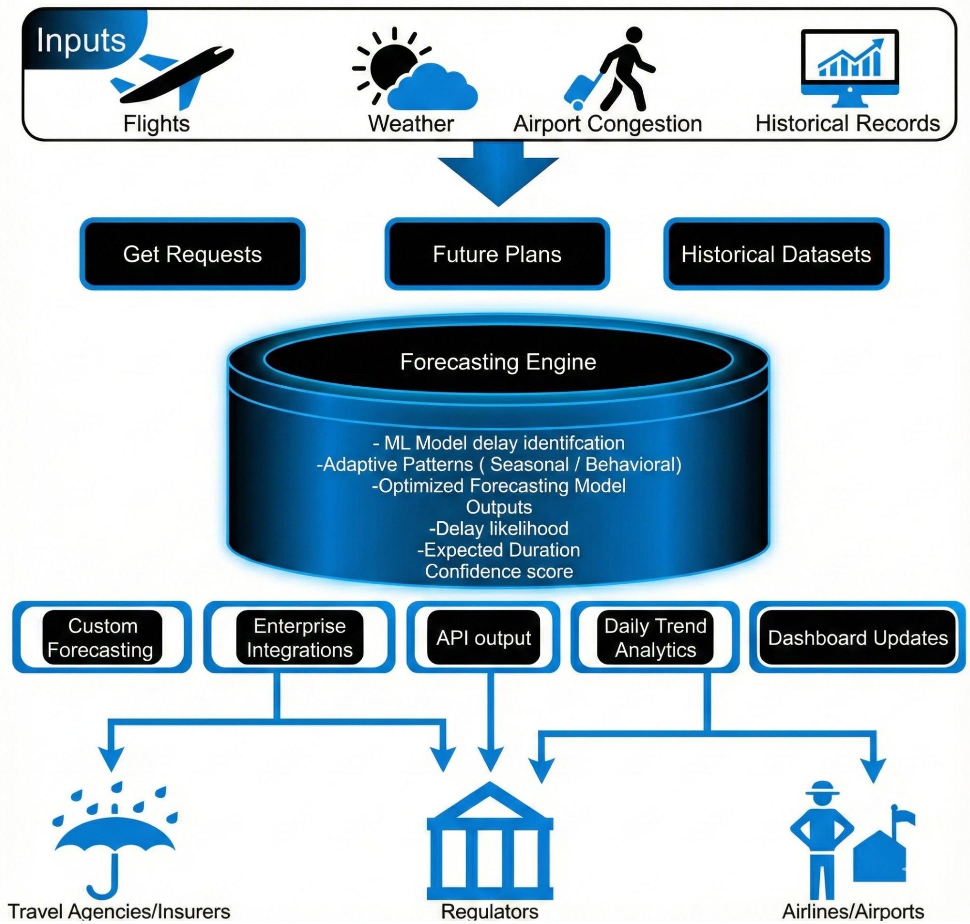
Room for Improvement

Since models are univariate, they only rely on historical values. Regional aggregation also hides micro-patterns from individual airports

- Univariate-only models (no weather, traffic, schedules)
- Aggregated regional data hides airport-specific dynamics
- Structural shocks remain challenging

Future Roadmap

- Adding real time live data for weather, congestion, etc.
- Building additional forecasting capabilities ex. duration of delay
- Multi platform integration support
- Refactoring for personal, traveller level use.



Lessons Learned

Lesson learned?

- Seasonality affect strongly to flight delays
- The 2020 COVID strongly impact the prediction

Anything surprising?

- The size of the performance jump when switching from non-seasonal to seasonal models was larger than expected.
- ETS models performed far worse than anticipated in regions with irregular fluctuations.

Was anything especially hard or easy?

- Hard: Managing multiple cutoff windows and ensuring consistent evaluation across all regions took the most time.

If doing this again, we want to explore more models including Prophet or Transformer-based models, and perform deeper feature engineering.

Conclusion & Recommendations

Delays look very different across the country. The South East and West see the biggest swings, while the North East and South West stay more steady. All regions follow the same seasonal pattern, with summer peaks and a visible drop in early 2020.

Our forecasts capture these patterns best when each region gets its own model. Seasonal Auto ARIMA performs the strongest overall, especially in high volatility areas. This approach helps pinpoint where delay risk rises and supports smarter staffing and planning.

Recommendations:

- Treat each region separately.
- Keep a close watch on the South East and West since they spike first.
- Use the forecasts to lock in staffing and gate plans early.
- Get ahead of summer because that is where delays jump.
- Use the steady regions as a baseline to spot unusual shifts.

Thank you.